INCH-POUND

MIL-I-24768/3
25 February 1992

SUPERSEDING
MIL-P-18177C
25 May 1960

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, GLASS-CLOTH, EPOXY-RESIN (GEB)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of plies or layers of glass cloth or nonwoven parallel aligned fibers impregnated and bonded with an epoxy-resin compound or binder, oriented and processed to conform to this specification.

Intended use: The laminated material covered by this specification is intended for use in electronics applications requiring moisture resistance and high mechanical strength. The laminate material GEB is not intended for applications requiring arc flame resistance.

Forms: Sheets are the only recommended form of laminated material covered by this specification.

Temperature indice: The laminated material covered by this specification has a mechanical strength test index of 160 and an electrical strength test index of 140 for thicknesses up to 0.062 inch, and a mechanical strength test index of 180 and an electrical strength test index of 170 for thicknesses greater than 0.062 inch.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

AMSC N/A FSC 5970

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TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)
M24768/3-S-1	0.010	0.002
M24768/3-S-2	0.015	0.003
M24768/3-S-3	0.020	0.004
M24768/3-S-4	0.025	0.005
M24768/3-S-5	0.031	0.0065
M24768/3-S-6	0.047	0.0075
M24768/3-S-7	0.062	0.0075
M24768/3-S-8	0.094	0.009
M24768/3-S-9	0.125	0.012
M24768/3-S-10	0.156	0.015
M24768/3-S-11	0.188	0.019
M24768/3-S-12	0.219	0.021
M24768/3-S-13	0.250	0.022
M24768/3-S-14	0.312	0.026
M24768/3-S-15	0.375	0.030
M24768/3-S-16	0.438	0.033
M24768/3-S-17	0.500	0.036
M24768/3-S-18	0.625	0.040
M24768/3-S-19	0.750	0.043
M24768/3-S-20	0.875	0.046
M24768/3-S-21	1.000	0.049
M24768/3-S-22	1.125	0.053
M24768/3-S-23	1.250	0.055
M24768/3-S-24	1.375	0.058
M24768/3-S-25	1.500	0.061
M24768/3-S-26	1.625	0.064
M24768/3-S-27	1.750	0.067
M24768/3-S-28	1.875	0.070
M24768/3-S-29	2.000	0.073
M24768/3-S-30	2.125	0.079
M24768/3-S-31	2.500	0.085
M24768/3-S-32	2.750	0.091
M24768/3-S-33	3.000	0.097
M24768/3-S-34	3.500	0.110
M24768/3-S-35	4.000	0.122
M24768/3-S-36	4.500	0.134
M24768/3-S-37	5.000	0.142
M24768/3-S-38	5.500	0.156
M24768/3-S-39	6.000	0.172
M24768/3-S-40	6.500	0.180
M24768/3-S-41	7.000	0.195
M24768/3-S-42	7.500	0.210
M24768/3-S-43	8.000	0.216
M24768/3-S-44	8.500	0.232
M24768/3-S-45	9.000	0.240
M24768/3-S-46	9.500	0.255
M24768/3-S-47	10.000	0.270
M24768/3-S-48	—[1]	—[2]

See footnotes on next page.

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[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/3-S-48-0.8125.
[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

Performance requirements: Performance requirements for laminated sheets shall be as specified in table II.

The following performance requirements do not apply to this plastic laminate:

a. Dielectric strength
b. Specific gravity
c. Compressive strength
d. Tracking resistant
e. Silicone content
f. Flammability.

Testing: The laminated material covered by this specification shall be tested under qualification inspection requirements.

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TABLE II. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Dielectric breakdown voltage[4]	A D-48/50	4.6.4	Min. kV	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	
Dielectric constant[2]	A D-24/23 D-48/50	4.6.6	Max.	5.2 5.4	5.2 5.4	5.2 5.4	5.2 5.4 5.8	5.2 5.4	5.2 5.4	5.2 5.4	5.5 5.7	5.5 5.7	5.7
Dissipation factor	A D-24/23 D-48/50	4.6.6	Max.	0.025 0.035	0.025 0.035	0.025 0.035	0.025 0.035 0.045	0.025 0.035	0.025 0.035	0.025 0.035	0.025 0.035	0.025 0.035	0.025 0.035
Volume resistivity	C-96/35/90	4.6.7	Min. meghom cm				10^5						
Surface resistance	C-96/35/90	4.6.8	Min. megohm				1,000						
Arc resistance[4]	D-48/50	4.6.9	Min. seconds										
Tracking resistance	A	4.6.10	Min. minutes										
Impact strength[4] Lengthwise: Crosswise:	E-48/50	4.6.11	Min. ft – lbs. per inch				7.0 5.5	7.0 5.5	7.0 5.5	7.0 5.5	7.0 5.5	7.0 5.5	7.0 5.5

See footnotes at the end of table.

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TABLE II. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Flexural strength			Min.										
Lengthwise:	A	4.6.12	psi	60,000	60,000	60,000	55,000	55,000	55,000	45,000	40,000	40,000	40,000
Crosswise:				50,000	50,000	50,000	45,000	45,000	45,000	35,000	30,000	30,000	30,000
Thermal endurance													
Flexural[3] strength			Min. psi										
Lengthwise:	E-1/150: T-150	4.6.13		30,000	30,000	30,000	30,000	30,000	30,000				
Bonding strength[4]	A	4.6.15	Min.							1,600	1,600	1,600	1,600
	D-48/50		pounds							1,500	1,500	1,500	1,500
Water absorption	D_1-24/23	4.6.17	Max. percent	0.50	0.25	0.20	0.15	0.15	0.13	0.10	0.10	0.10	0.10
Silicone content	E-168/185	4.6.18	Max. ppm										
Flame resistance													
Ignition time:	A	4.6.19	Min. sec.										
Burning time:			Max. sec.										

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.
[2]Sheets with a thickness of 0.501 inch or more shall have a value corresponding to the next higher value.
[3]Flexural strength shall be not less than 30,000 psi for the specified thickness.
[3]Maximum thickness tested shall be 2.000 inches.

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Army – ER
Navy – SH
Air Force – 20

Review activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User activities:
Army – ME
Navy – MC, AS

Preparing activity:
Navy – SH
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